Superconformal Index Of 4d $\mathcal{N} = 2$ SCFTs From Chiral Algebras And Monodromy Operator

Ranveer Kumar Singh

NHETC
Department of Physics and Astronomy
Rutgers University

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Outline

1. Superconformal Index Of 4d $\mathcal{N} = 2$ SCFTs
2. Schur Index Using SCFT/VOA
3. Macdonald Index From VOA
4. Schur Index Using Monodromy Operator
5. Macdonald Index From Monodromy Operator
6. Future Directions

Superconformal Index Of 4d $\mathcal{N} = 2$ SCFTs

Definition

4d $\mathcal{N}=2$ superconformal algebra:

- Lorentz (Euclidean): $\mathrm{SU}(2)\times\mathrm{SU}(2)$, Cartan generators $(j,\bar{j})$.
- Scaling Cartan generator E.
- R-symmetry $\mathrm{SU}(2)\times\mathrm{U}(1)$, Cartan generators (R,r).
- Supercharges: Poincaré $Q^I_\alpha, \widetilde{Q}^{\dot{\alpha}}_J$, Conformal $(Q^I_\alpha)^\dagger, (\widetilde{Q}^{\dot{\alpha}}_J)^\dagger$.
- Special conformal transformations: generators annihilate conformal primaries.

Superconformal index: consider the theory on $S^3\times S^1$ and define

$$\mathcal{I}_{\mathrm{SCI}}(p,q,t)=\mathrm{Tr}_{\mathscr{H}(S^3)}\left[(-1)^F p^{j-\bar{j}-r}q^{j+\bar{j}-r}t^{R+r}\right].$$

Kinney-Maldacena-Minwalla-Raju, Römelsberger

Can also allow flavor fugacities.
Only $\frac{1}{8}$-BPS states contribute.

Macdonald and Schur Index

Macdonald limit: $p \to 0$:

$$\mathcal{I}_M(q,t) = \mathrm{Tr}_{\mathscr{H}(S^3)}\left[(-1)^F q^{E-2R-r} t^{R+r}\right].$$

Counts $\frac{1}{4}$-BPS states. Gadde-Rastelli-Razamat-Yan

Schur limit: $t \to q$

$$\mathcal{I}_S(q) = \mathrm{Tr}_{\mathscr{H}(S^3)}\left[(-1)^F q^{E-R}\right].$$

Also counts $\frac{1}{4}$-BPS states but with less information.

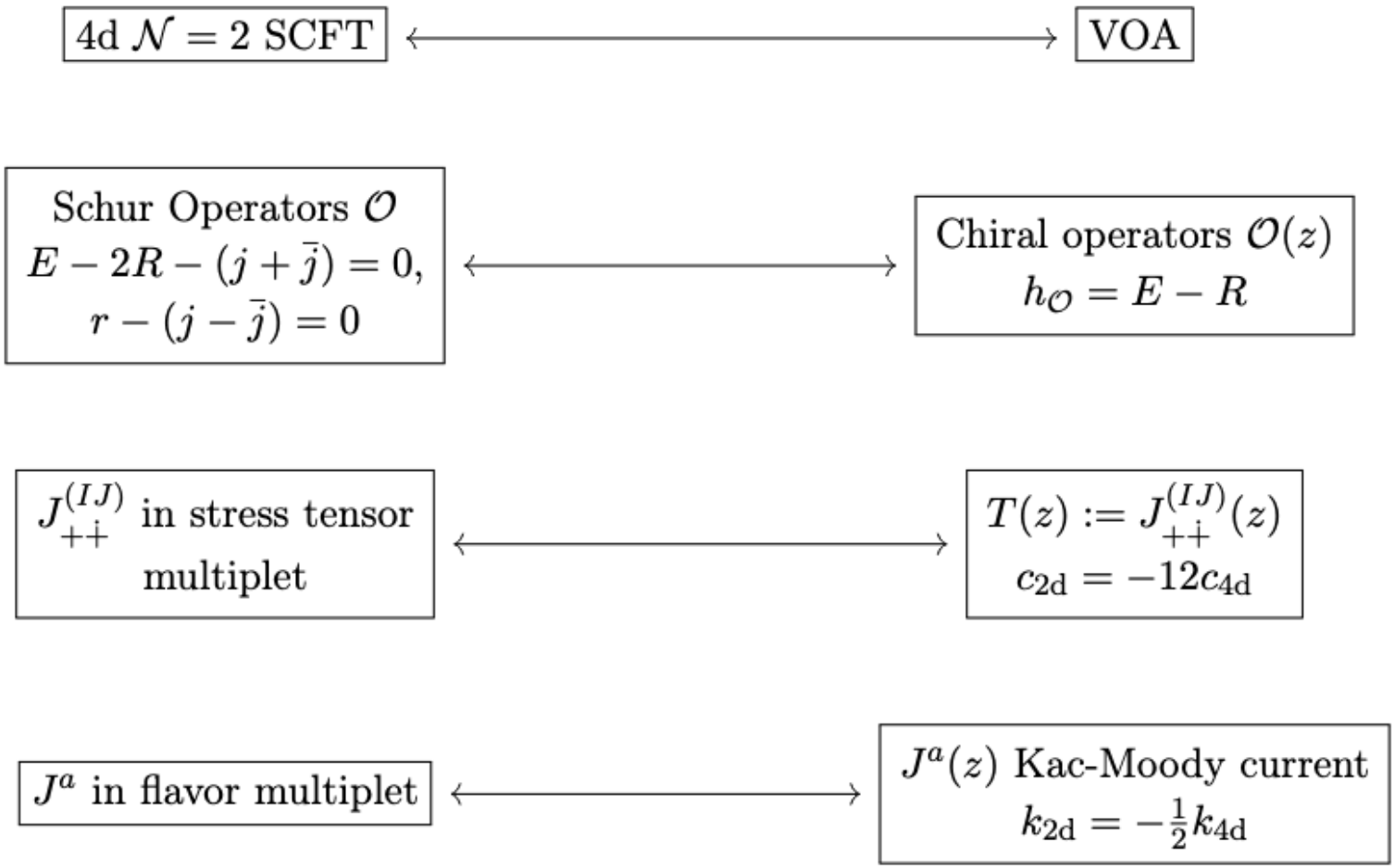
Gadde-Rastelli-Razamat-Yan

$$\mathcal{I}_S(q) = \lim_{t\to q} \mathcal{I}_M(q,t)\ .$$

Various indirect ways of computing the index: IR fixed point of Lagrangian theories, SCFT/VOA correspondence, Monodromy operator, class S techniques, q-deformed Yang-Mills...

Schur Index Using SCFT/VOA

SCFT/VOA Correspondence



SCFT/VOA Correspondence implies $\mathcal{I}_{\mathrm{S}}(q)=\chi_{\mathcal{V}}(q)$.

Example: Argyres-Douglas theories

- A class of non-Lagrangian theories labeled by two ADE Lie algebras $(\mathfrak{g}_1, \mathfrak{g}_2)$.

Cecotti-Neitzke-Vafa

- Construction using Type IIB compactified on singular Calabi-Yau.

Klemm-Lerche-Mayr-Vafa-Warner, Ooguri-Vafa

- Singularities on Coulomb branch of Lagrangian theories.

Argyres-Douglas, Argyres-Plesser-Seiberg-Witten

- IR fixed points of $\mathcal{N} = 1$ Lagrangian theories. Maruyoshi-Song
- These constructions give us ways of computing the SCI.

$$V_{(A_1, A_{2n})} = \mathrm{Vir}_{c_n}, \quad c_n = -\frac{2n(6n+5)}{2n+3}.$$

$$V_{(A_1, D_{2n+1})} = \widehat{\mathfrak{su}(2)}_{-\frac{4n}{2n+1}}, \quad c_n = -6n.$$

Beem-Rastelli

Schur Indices Of (A_1, A_{2n}) And (A_1, D_{2n+1}) AD Theories

$$\mathcal{I}_{\rm S}^{(A_1,A_{2n})}(q) = \sum_{N_1\geq N_2\geq\cdots\geq N_n\geq 0} \frac{q^{N_1^2+\cdots+N_n^2+N_1+N_2+\cdots+N_n}}{(q;q)_{N_1-N_2}\cdots(q;q)_{N_{n-1}-N_n}(q;q)_{N_n}},$$

where $(a,q)_n = \prod_{k=0}^{n-1}(1-aq^k)$.

$$\mathcal{I}_{\rm S}^{(A_1,D_{2n+1})}(q) = \left(\frac{(q^{2n+1};q^{2n+1})_\infty}{(q;q)_\infty}\right)^3, \quad (a;q)_\infty = \prod_{k=0}^{\infty}(1-aq^k).$$

Buican-Nishinaka

Matches with the vacuum character of the corresponding VOAs.

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Question: Can we get the Macdonald index from VOA?

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Question: Can we get the Macdonald index from VOA?

Partial answer: Refined character of the VOA

Song, Watanabe-Zhu

Macdonald Index From VOA

Strongly finitely generated vertex operator algebras

Vertex operator (super)algebra is a

- $(\frac{1}{2})\mathbb{Z}$-graded vector space $V = \bigoplus_{k\in(\frac{1}{2})\mathbb{Z}} V_{(k)}$.
- a vertex operator map $Y(v,z) = \sum_{n\in\mathbb{Z}} v_n z^{-n-1}$, $v_n \in \mathrm{End}(V)$.
- A conformal vector $\omega \in V_{(2)}$, such that L_n defined by $T(z) := Y(\omega, z) = \sum_{n\in\mathbb{Z}} L_n z^{-n-2}$ satisfies the Virasoro algebra with central charge c.
- Vacuum vector $|0\rangle \in V_{(0)}$ such that $Y(|0\rangle, z) = 1, \lim_{z\to 0} Y(v,z)|0\rangle = v$.
- Axioms: OPE, locality, commutativity...

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A VOA is called *strongly finitely generated* if there exists $u^1, \dots, u^N$ such that

$$V = \mathrm{Span}\{u^{i_1}_{-n_1} u^{i_2}_{-n_2} \cdots u^{i_k}_{-n_k}|0\rangle : n_1, n_2, \dots, n_k > 0, k \geq 0\}/\{\text{null vectors}\}$$

$u^1, \dots, u^N$ are called *strong generators*.

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$u^1, \dots, u^N$ are called *strong generators*. Affine and Virasoro VOAs are strongly finitely generated.

Zhu algebra

- Define

$$\mathcal{A}(V) := V/C_2(V), \quad C_2(V) := \mathrm{Span}\{a_{-2} \cdot b : a, b \in V\}.$$

- $\mathcal{A}(V)$ is a commutative associative algebra called *Zhu algebra*. Zhu
- It can be proved that for a strongly finitely generated VOA with N strong generators Xie-Yan, Andrews-Banerjee-**RKS**-Tao-Bhargava

$$\mathcal{A}(V) \cong \mathbb{C}[x^{(1)}, \ldots, x^{(N)}]/I, \quad (u^i \mapsto x^{(i)})$$

  where I is the ideal generated by null vector relations.
- Example:

$$\mathcal{A}\left(\mathrm{Vir}_{-\frac{2n(6n+5)}{2n+3}}\right) \cong \mathbb{C}[x]/\langle x^{n+1} \rangle.$$

$$\mathcal{A}\left(\widehat{\mathfrak{su}(2)}_{-\frac{4n}{2n+1}}\right) \cong \mathbb{C}[x, y, z, j]/\langle f_1, f_2, f_3, f_4, f_5 \rangle,$$

  where Beem-Rastelli

$f_1 = xz - y^2 - j, f_2 = j^{n+1}x, f_3 = j^{n+1}y, f_4 = j^{n+1}z, f_5 = j^{n+1}.$

Arc space of a polynomial ring

Let $R = \mathbb{C}[x^{(1)}, \ldots, x^{(N)}]/\langle f_1, \ldots, f_K \rangle$ be a polynomial ring. Assign weights $d_i \geq 1$ to $x^{(i)}$ and write formal power series

$$x^{(i)}(s) = \sum_{n=d_i-1}^{\infty} x_n^{(i)} s^n .$$

The n-th jet space of R is defined by

$$\begin{aligned} R_n &:= \mathbb{C}[x_{d_1-1}^{(1)}, \ldots, x_n^{(1)}, x_{d_2-1}^{(2)}, \ldots, x_n^{(2)}, \ldots, x_{d_N-1}^{(N)}, \ldots, x_n^{(N)}]/I_n \ , \\ I_n &:= \left\langle f_1(\vec{x}(s)), \ldots, f_K(\vec{x}(s)) \equiv 0 \bmod s^{n+1} \right\rangle \ . \end{aligned}$$

The n-th Jet scheme is defined as

$$X_n(R) := \operatorname{Spec} R_n \ .$$

The *arc space* $X_\infty(R)$ of R is defined to be

$$X_\infty(R) = \varprojlim X_n(R) \cong \operatorname{Spec} R_\infty \ .$$

Hilbert series of arc space

Assign bigrading $\mathrm{wt}(x_j^{(i)}) = (j, w(x^{(i)}))$ and[1] define

$$H_{R_\infty}(q,t) := \sum_{\ell,m\geq 0} \dim(R_\infty)_{\ell,m} q^\ell t^m \ .$$

Example: $R = \mathbb{C}[x]/\langle x^2\rangle$. Choosing $w(x) = 1$, the Hilbert series is given by

$$H_{R_\infty}(q,t) = \sum_{m\geq 0} t^m \frac{q^{m^2}}{(q;q)_m}.$$

Bruschek-Mourtada-Schepers, Bai-Gorsky-Kivinen

This agrees with the Macdonald index of (A_1, A_2) theory!!

Buican-Nishinaka-Bhargava

[1] Warning: R-filtration!!

Illustration: $R = \mathbb{C}[x, y, z, j]/\langle xz - y^2 - j, jx, jy, jz, j^2\rangle$

Choose $\mathrm{wt}(x, y, z) = 1, \mathrm{wt}(j) = 2, w(x, y, z, j) = 1$. We have the formal series

$$x(s) = \sum_{n=0}^{\infty} x_n s^n\,, \quad y(s) = \sum_{n=0}^{\infty} y_n s^n\,, \quad z(s) = \sum_{n=0}^{\infty} z_n s^n\,, \quad j(s) = \sum_{n=1}^{\infty} j_n s^n\,.$$

n-th jet space is

$$\begin{aligned} R_n &:= \mathbb{C}[x_0, \ldots, x_n, y_0, \ldots, y_n, z_0, \ldots, z_n, j_1, \ldots, j_n]/I_n, \\ I_n &:= \langle f_i(x(s), y(s), z(s), j(s)) \equiv 0 \bmod s^{n+1}\rangle. \end{aligned}$$

For example:

$$\begin{aligned} I_0 &= \langle x_0 z_0 - y_0^2\rangle\,, \\ I_1 &= \langle x_0 z_0 - y_0^2, x_0 z_1 - 2y_0 y_1 + z_0 x_1 - j_1, x_0 j_1, y_0 j_1, z_0 j_1\rangle \end{aligned}$$

Illustration continued...

$$\left.\begin{array}{l}(R_\infty)_{0,0} : 1 \\ (R_\infty)_{0,1} : x_0, y_0, z_0 \\ (R_\infty)_{n,1} : x_n, y_n, z_n, j_n\end{array}\right\} \Longrightarrow H_{R_\infty}(q,t) = 1 + \left(3 + 4\sum_{n=1}^{\infty} q^n\right) t + O(t^2).$$

Illustration continued...

$$\left.\begin{array}{l}(R_\infty)_{0,0}:1\\(R_\infty)_{0,1}:x_0,y_0,z_0\\(R_\infty)_{n,1}:x_n,y_n,z_n,j_n\end{array}\right\}\implies H_{R_\infty}(q,t)=1+\left(3+4\sum_{n=1}^{\infty}q^n\right)t+O(t^2).$$

Next

$$(R_\infty)_{0,2}:\mathbb{C}[x_0^2,y_0^2,z_0^2,x_0y_0,x_0z_0,y_0z_0]/\langle x_0z_0-y_0^2\rangle\implies\dim(R_\infty)_{0,2}=5.$$

Counting similarly, $\dim(R_\infty)_{1,2}=8, \dim(R_\infty)_{2,2}=17$.
Systematic method: using Gröbner basis algorithm, we found a general formula for the Hilbert series. Can be evaluated using Mathematica.

$$H_{R_\infty}(q,t)=\sum_{m=0}^{\infty}\frac{t^m}{(q;q)_m}\sum_{k=0}^{2m}\frac{(q;q)_{2m}}{(q;q)_k(q;q)_{2m-k}}.$$ Andrews-Banerjee-**RKS**-Tao-Bhargava

Matches with the Macdonald index of (A_1,D_3) theory!!

Conjecture

Conjecture: Macdonald index from VOA Andrews-Banerjee-**RKS**-Tao-Bhargava

Let V_T be the 2d VOA corresponding to a 4d $\mathcal{N}=2$ SCFT T and let $\mathcal{A}(V_T) \cong \mathbb{C}[x^{(1)}, \ldots, x^{(N)}]/I$ be its Zhu algebra. Assign the grading $\mathrm{wt}(x^{(i)}) = d_i$, where d_i is the conformal dimension of strong generator u^i. Then there exists $w(x^{(i)})$ such that

$$\mathcal{I}^T_{\mathrm{M}}(q,t) = H_{\mathcal{A}(V_T)_\infty}(q,t).$$

Checked it for (A_1, A_{2n}) and (A_1, D_{2n+1}) theories.
Equating the Schur limit with the known Schur index gives new q-series identities.

- (A_1, A_{2n}) theory gives Rogers-Ramanujan identities.

Buican-Nishinaka-Bhargava

- We proved new identities coming from (A_1, D_{2n+1}) theory.

$$(A_1, D_3): \left(\frac{(q^3;q^3)_\infty}{(q;q)_\infty}\right)^3 = \sum_{m=0}^{\infty} \frac{q^m}{(q;q)_m} \sum_{k=0}^{2m} \frac{(q;q)_{2m}}{(q;q)_k (q;q)_{2m-k}}.$$

Schur Index From Monodromy Operator

Coulomb branch geometry basics

4d $\mathcal{N} = 2$ theories generically have moduli space of vacua: Coulomb branch, Higgs branch and mixed branch.

- Coulomb branch $\mathcal{B}$ parametrized by u. Rank $= \dim_{\mathbb{C}}(\mathcal{B}) = r$.
- Charge lattice: Γ_u with antisymmetric pairing $\langle \cdot, \cdot \rangle$.
- Central charge function $\mathcal{Z} : \Gamma_u \to \mathbb{C}$.
- BPS indices $\Omega : \Gamma \to \mathbb{Z}$ which obeys *wall crossing formula*.

BPS Index

Restrict to the sublattice $\Gamma^{\rm BPS}$ of BPS charges[2].

- BPS particles are in irreducible representations of $\mathrm{SU}(2)_L \times \mathrm{SU}(2)_R$, where $\mathrm{SU}(2)_L$ is the little group of the Lorentz group.
- Let $\mathscr{H}^\gamma$ be the Hilbert space of BPS states with charge γ. We can factor out the center-of-mass dof to write

$$\mathscr{H}^\gamma = [(\mathbf{2},\mathbf{1}) \oplus (\mathbf{1},\mathbf{2})] \otimes \mathscr{H}^\gamma_{\rm int}.$$

- Define the Protected Spin Character

$$\Omega_\gamma(y) = \mathrm{Tr}_{\mathscr{H}^\gamma_{\rm int}} (-1)^R y^{J+R} = \sum_{n\in\mathbb{Z}} \Omega_n(\gamma) y^n, \quad \Omega_n(\gamma) \in \mathbb{Z}.$$

  where J is the Cartan generator of $\mathrm{SU}(2)_L$. Gaiotto-Moore-Neitzke

- $\Omega_\gamma(y)$ jumps across walls of marginal stability and obeys a wall crossing formula. Denef, Denef-Moore, Gaiotto-Moore-Neitzke

[2] Dropped the u-dependence.

Monodromy operator

Quantum torus algebra

$$\mathcal{A} = \mathrm{Span}\{X_\gamma : \gamma \in \Gamma^{\mathrm{BPS}}\}$$

$$X_\gamma X_{\gamma'} = q^{\frac{\langle\gamma,\gamma'\rangle}{2}} X_{\gamma+\gamma'} = q^{\langle\gamma,\gamma'\rangle} X_{\gamma'} X_\gamma.$$

Define the quantum dilogarithm

$$E_q(z) := \prod_{i=0}^{\infty} \left(1 + q^{i+\frac{1}{2}} z\right)^{-1} = \sum_{n=0}^{\infty} \frac{(-q^{\frac{1}{2}} z)^n}{(q;q)_n}.$$

The monodromy operator is defined as

$$\mathcal{O}(q) := \prod_{\gamma\in\Gamma^{\mathrm{BPS}}}^{\curvearrowright} U_\gamma, \quad U_\gamma = \prod_{n\in\mathbb{Z}} E_q\left((-1)^n q^{n/2} X_\gamma\right)^{(-1)^n \Omega_n(\gamma)},$$

where the product is taken in the increasing order of $\mathrm{Arg}(\mathcal{Z}(\gamma))$ with respect to a reference angle θ.

Cordova-Shao Conjecture

A charge $\gamma \in \Gamma^{\mathrm{BPS}}$ is called a flavor charge if $\langle \gamma, \Gamma^{\mathrm{BPS}} \rangle = 0$. Any flavor charge can be expanded in a basis $\gamma_f = \sum_{i=1}^{N_f} f_i(\gamma)\gamma_{f_i}$. Define

$$\mathrm{Tr}[X_\gamma] = \begin{cases} \prod_i \mathrm{Tr}\left[X_{\gamma_{f_i}}\right]^{f_i(\gamma)} & \langle \gamma, \Gamma^{\mathrm{BPS}} \rangle = 0 \ , \\ 0 & \text{else} \ , \end{cases} \ .$$

Cordova-Shao Conjecture

The Schur index of a 4d $\mathcal{N} = 2$ SCFT of rank r is given by

$$\mathcal{I}_{\mathrm{S}}(q, z_1, \ldots, z_{N_f}) := (q;q)_\infty^{2r} \, \mathrm{Tr}\, \mathcal{O}(q),$$

where z_i are the flavor fugacities and related to $\mathrm{Tr}\left[X_{\gamma_{f_i}}\right]$.

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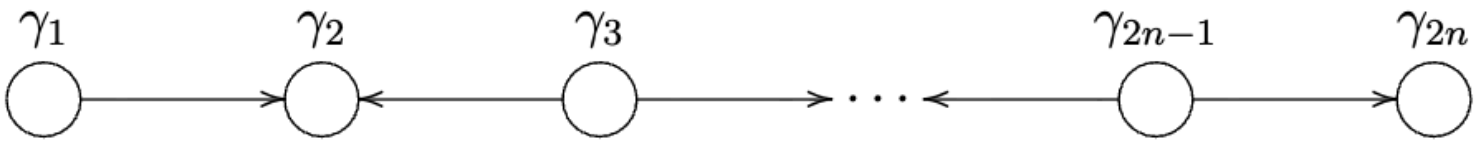
where z_i are the flavor fugacities and related to $\mathrm{Tr}\left[X_{\gamma_{f_i}}\right]$.

Can we compute Macdonald index from trace of monodromy operator?

Macdonald Index From Monodromy Operator

BPS quivers of 4d $\mathcal{N} = 2$ theories

The BPS spectrum of some theories in a given chamber is captured by its BPS quiver. Prominent example: $(\mathfrak{g}_1, \mathfrak{g}_2)$ theories. $(A_1, \mathfrak{g})$ theories admit source/sink chamber.
Example: (A_1, A_{2n}) theory.



Cecotti-Neitzke-Vafa

- $\mathrm{Arg}(\mathcal{Z}(\gamma)) > \mathrm{Arg}(\mathcal{Z}(\gamma'))$ for $\gamma \in$ sink and $\gamma' \in$ source.
- $[X_\gamma, X_{\gamma'}] = 0$, for $\gamma, \gamma' \in$ sources, or $\gamma, \gamma' \in$ sinks.
- $\langle \gamma_i, \gamma_j \rangle$ determined by the number of arrows between i, j.

$$\mathcal{O}(q) = \prod_{i=1}^{n} E_q(X_{-\gamma_{2i-1}}) \prod_{i=1}^{n} E_q(X_{-\gamma_{2i}}) \prod_{i=1}^{n} E_q(X_{\gamma_{2i-1}}) \prod_{i=1}^{n} E_q(X_{\gamma_{2i}}) \ .$$

Refined monodromy operator

Define

$$E_{q,T}(X_\gamma) = \sum_{n=0}^{\infty} \frac{(-(qT)^{\frac{1}{2}} X_\gamma)^n}{(q;q)_n}, \quad \widetilde{E}_{q,T}(X_\gamma) = \sum_{n=0}^{\infty} \frac{(-q^{\frac{1}{2}} X_\gamma)^n}{(qT;q)_n}.$$

Prescription:

$$E_q(X_{-\gamma_{\text{source}}}) \to E_{q,T}(X_{-\gamma_{\text{source}}}), \quad E_q(X_{-\gamma_{\text{sink}}}) \to \widetilde{E}_{q,T}(X_{-\gamma_{\text{sink}}})$$
$$E_q(X_{\gamma_{\text{source}}}) \to E_{q,T}(X_{\gamma_{\text{source}}}), \quad E_q(X_{\gamma_{\text{sink}}}) \to E_q(X_{\gamma_{\text{sink}}}).$$

If there is a source with valency > 2 then exchange the roles of sources and sinks and apply the above prescription.

Conjecture Andrews-Banerjee-**RKS**-Tao

For a 4d $\mathcal{N} = 2$ SCFT with source/sink chamber, the Macdonald index is given by

$$\mathcal{I}_{\text{M}}(q,T) = (q;q)_\infty^r (qT;q)_\infty^r \text{Tr}[\mathcal{O}(q,T)].$$

$qT = t$ and Schur limit is $T \to 1$. Since $\mathcal{O}(q,1) = \mathcal{O}(q)$, the Schur limit gives the Schur index.

Computing the trace for (A_1, A_{2n}) theory

For (A_1, A_{2n}), we get

$$\mathcal{O}(q,T) = \prod_{i=1}^{n} E_{q,T}(X_{-\gamma_{2i-1}}) \prod_{i=1}^{n} \widetilde{E}_{q,T}(X_{-\gamma_{2i}}) \prod_{i=1}^{n} E_{q,T}(X_{\gamma_{2i-1}}) \prod_{i=1}^{n} E_{q}(X_{\gamma_{2i}}) \,.$$

Replace the definition of $E_{q,T}, \widetilde{E}_{q,T}$.

$$\mathcal{O}(q,T) = \sum_{\substack{\ell_1,\ldots,\ell_{2n}\geq 0 \\ k_1,\ldots,k_{2n}\geq 0}} \frac{(-1)^{\sum_{i=1}^{2n}(k_i+\ell_i)} q^{\frac{1}{2}\sum_{i=1}^{2n}(k_i+\ell_i)} T^{\frac{1}{2}\sum_{i=1}^{n}(\ell_{2i-1}+k_{2i-1})}}{\prod_{i=1}^{n}(q)_{k_{2i-1}}(q)_{\ell_{2i-1}} \prod_{i=1}^{n}(q)_{k_{2i}}(qT)_{\ell_{2i}}}$$
$$\times \prod_{i=1}^{n} X_{-\gamma_{2i-1}}^{\ell_{2i-1}} \prod_{i=1}^{n} X_{-\gamma_{2i}}^{\ell_{2i}} \prod_{i=1}^{n} X_{\gamma_{2i-1}}^{k_{2i-1}} \prod_{i=1}^{n} X_{\gamma_{2i}}^{k_{2i}} \,.$$

Shortened notation $(a)_n = (a;q)_n$.

Computation of trace for (A_1, A_{2n}) theory continued...

Use quantum torus algebra to write

$$\prod_{i=1}^{n} X_{-\gamma_{2i-1}}^{\ell_{2i-1}} \prod_{i=1}^{n} X_{-\gamma_{2i}}^{\ell_{2i}} \prod_{i=1}^{n} X_{\gamma_{2i-1}}^{k_{2i-1}} \prod_{i=1}^{n} X_{\gamma_{2i}}^{k_{2i}}$$

$$= q^{-\sum_{i,j=1}^{n} k_{2j-1}\ell_{2i} C^{A_{2n}}_{2i,2j-1}} \prod_{i=1}^{n} X_{\gamma_{2i-1}}^{k_{2i-1}-\ell_{2i-1}} \prod_{i=1}^{n} X_{\gamma_{2i}}^{k_{2i}-\ell_{2i}}$$

where $C_{ij}^{A_{2n}}$ is the Cartan matrix of A_{2n}. Use quantum torus algebra again

$$\mathrm{Tr}\left[\prod_{i=1}^{n} X_{\gamma_{2i-1}}^{k_{2i-1}-\ell_{2i-1}} \prod_{i=1}^{n} X_{\gamma_{2i}}^{k_{2i}-\ell_{2i}}\right] = q^{\#}\mathrm{Tr}\left[X_{\sum_{i=1}^{2n}(k_i-\ell_i)\gamma_i}\right] = \prod_{i=1}^{2n} \delta_{k_i,\ell_i}\,.$$

Last step follows from absence of flavor charges. After some simplification, our conjecture gives

$$\mathcal{I}_{\mathrm{M}}^{(A_1,A_{2n})}(q,T) = (q)_\infty^n (qT)_\infty^n \sum_{k_1,\ldots,k_{2n}\geq 0} \frac{q^{\sum_{i=1}^{2n} k_i + \frac{1}{2}\sum_{i=1}^{2n-1} k_i k_{i+1}} T^{\sum_{i=1}^{n} k_{2i-1}}}{\prod_{i=1}^{n} (q)_{k_{2i-1}}^2 \prod_{i=1}^{n} (q)_{k_{2i}} (qT)_{k_{2i}}}\,.$$

New identity

Equating with the known Macdonald index gives

$$(q)_\infty^n (Tq)_\infty^n \sum_{k_1,\ldots,k_{2n}\geq 0} \frac{T^{k_1+k_3+\ldots+k_{2n-1}} q^{k_1k_2+k_2k_3+\ldots+k_{2n-1}k_{2n}+k_1+\ldots+k_{2n}}}{\prod_{i=1}^n (q)_{k_{2i-1}}^2 \prod_{i=1}^n (q)_{k_{2i}} (Tq)_{k_{2i}}}$$

$$= \sum_{\ell_1,\ldots,\ell_n\geq 0} \frac{T^{\ell_1+\ldots+\ell_n} q^{\ell_1^2+\ldots+\ell_n^2+\ell_1+\ldots+\ell_n}}{(q)_{\ell_1-\ell_2}(q)_{\ell_2-\ell_3}\cdots(q)_{\ell_{n-1}-\ell_n}(q)_{\ell_n}}.$$

Can be proved rigorously.

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New identity

Equating with the known Macdonald index gives

$$(q)_\infty^n (Tq)_\infty^n \sum_{k_1,\ldots,k_{2n}\geq 0} \frac{T^{k_1+k_3+\ldots+k_{2n-1}} q^{k_1k_2+k_2k_3+\ldots+k_{2n-1}k_{2n}+k_1+\ldots+k_{2n}}}{\prod_{i=1}^n (q)_{k_{2i-1}}^2 \prod_{i=1}^n (q)_{k_{2i}} (Tq)_{k_{2i}}}$$

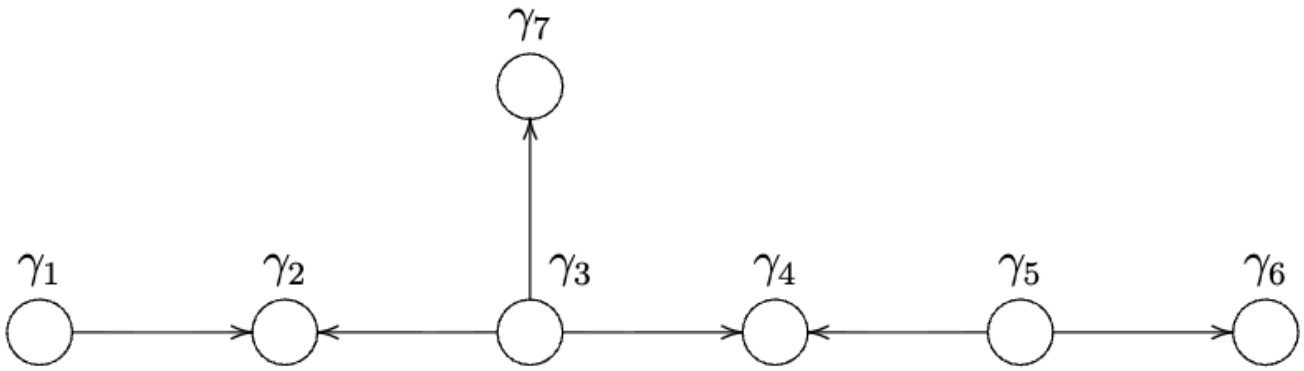
$$= \sum_{\ell_1,\ldots,\ell_n\geq 0} \frac{T^{\ell_1+\ldots+\ell_n} q^{\ell_1^2+\ldots+\ell_n^2+\ell_1+\ldots+\ell_n}}{(q)_{\ell_1-\ell_2}(q)_{\ell_2-\ell_3}\cdots(q)_{\ell_{n-1}-\ell_n}(q)_{\ell_n}}.$$

Can be proved rigorously. Andrews-Banerjee-**RKS**-Tao

Reproduces the Macdonald index of all $(A_1, \mathfrak{g})$ theories

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Macdonald index of (A_1, E_7) theory



Cecotti-Neitzke-Vafa

Source with valency > 2. With our prescription

$$\mathcal{O}(q,T) = \prod_{i=1}^{3} E_{q,T}(X_{-\gamma_{2i}}) \prod_{i=1}^{4} \widetilde{E}_{q,T}(X_{-\gamma_{2i-1}}) \prod_{i=1}^{3} E_{q,T}(X_{\gamma_{2i}}) \prod_{i=1}^{4} E_{q}(X_{\gamma_{2i-1}})$$

The theory has U(1) flavor symmetry. Let z be the flavor fugacity. Choose the flavor charge to be $\gamma_f = \gamma_4 - \gamma_6 - \gamma_7$ and $\mathrm{Tr}[X_{\gamma_f}] = -z$.

Macdonald index of (A_1, E_7) theory continued...

The Macdonald index is given by

$$\mathcal{I}_{\rm M}^{(A_1,E_7)}(q,T,z)$$

$$= (q)_\infty^3 (qT)_\infty^3 \sum_{\substack{\ell_1,\cdots,\ell_7\geq 0\\ k_1,\cdots,k_7\geq 0}} \frac{(-1)^{\sum_{i=1}^7 (k_i+\ell_i)} T^{\frac{1}{2}(k_7+\ell_7)+\frac{1}{2}\sum_{i=1}^3 (k_{2i}+\ell_{2i})}}{(q)_{k_7}(q)_{\ell_7}\prod_{i=1}^3 (qT)_{k_{2i-1}}(q)_{\ell_{2i-1}}}$$

$$\times \frac{q^{\frac{1}{2}\sum_{i=1}^7 (k_i+\ell_i)} q^{k_2(\ell_1+\ell_3)+k_4(\ell_3+\ell_5)+k_6\ell_5+k_7\ell_3}}{\prod_{i=1}^3 (q)_{k_{2i}}(q)_{\ell_{2i}}}$$

$$\times (-z)^{-\ell_7+k_7}\delta_{k_4+k_7,\ell_4+\ell_7}\delta_{k_6-k_7,\ell_6-\ell_7}\prod_{i\in\{1,2,3,5\}}\delta_{k_i,\ell_i}$$

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Limitations

- The VOA approach is incomplete because of R-filtration problem, i.e., determining the weights $w(x^{(i)})$. Song, Beem-Maneghelli-Rastelli
- The prescription from monodromy operator is only for theories with a source/sink chamber.
- The Cordova-Shao prescription works even for non-conformal Lagrangian theory. Our proposal does not seem to work for non-conformal theories.

Future directions

- Generalize the prescription to general 4d $\mathcal{N} = 2$ SCFTs.
- Proof using 3d TFTs? Gang-Kim-Stubbs, Kim-Kim-Song
- Generalization of wall-crossing formula?

Thank You